Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 55 – EM Wave Intensity

1) Electromagnetic waves consist of

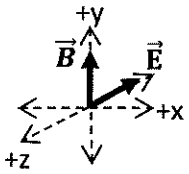
A) compressions and rarefactions of electromagnetic pulses.
(B) oscillating electric and magnetic fields.
C) particles of light energy.
D) high-frequency sound waves.

2) For an EM wave, how is the energy in the electric field compared to the energy in the magnetic field?

(A) The Electric field energy is the same as the energy in the magnetic field.
B) The Electric field energy is the greater than the energy in the magnetic field.
C) The Electric field energy is the less than the energy in the magnetic field.

3) At a certain instant in time, the electric field of an electromagnetic wave in free space points in the *-z* direction, and the magnetic field points in the *+y* direction. In what direction is this wave traveling?

(A) *+x* direction
B) *-x* direction
C) *+y* direction
D) *-z* direction
E) *+z* direction



$\vec{S} = \vec{E} \times \vec{B}$

right hand rule

4) Intensity of an EM wave is equal to

A) Power*area
B) Energy*area
(C) Power/area
D) Energy/area

5) If an EM wave has an Intensity of 500 W/m^2 and is incident on a solar cell with an area of 2 m^2, how much energy per second is incident on the solar cell?

A) 1,000 J/s
(B) 500 J/s
C) 250 J/s
D) 2,000 J/s

$I = \frac{P}{A}$ | $P = IA = 500\,W = 500\,J/s$

6) The value of the electric field for a certain type of electromagnetic wave is 570 N/C. What is the rms value of the magnetic field for that wave?

$|\vec{E}| = c|\vec{B}|$ | $|\vec{B}| = \frac{|\vec{E}|}{c} = \frac{570\,N/C}{3\times10^8\,T/s} = 1.9\times10^{-6}\,T$

7) The value of the magnetic field for a certain type of electromagnetic wave is 5×10^{-6} T. What is the rms value of the Electric field for that wave?

$|\vec{E}| = c\,|\vec{B}| = 1{,}500$ N/C

8) The value of the electric field for a certain type of electromagnetic wave is 570 N/C. What is the Intensity of the wave? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C²/Nm²)

$I = c\,\varepsilon_o\,E^2 = 863$ W/m²

9) The value of the magnetic field for a certain type of electromagnetic wave is 5×10^{-6} T. What is the Intensity of the wave? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C²/Nm²)

$I = \dfrac{cB^2}{\mu_0} = 5{,}970$ W/m²

10) The Intensity of an EM Wave is 30 W/m². What is the rms value of the magnetic field in the wave? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C²/Nm²)

$I = \dfrac{cB^2}{\mu_0} \quad | \quad B = \sqrt{\dfrac{\mu_0 I}{c}} = 3.54 \times 10^{-7}$ T

11) The Intensity of an EM Wave is 30 W/m². What is the peak value of the magnetic field in the wave? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C²/Nm²)

$B_{rms} = \sqrt{\dfrac{\mu_0 I}{c}} = 3.54 \times 10^{-7}$ T $\quad | \quad B_{rms} = \dfrac{B_p}{\sqrt{2}}$

$B_p = B_{rms}(\sqrt{2}) = 5 \times 10^{-7}$ T

12) The Intensity of an EM Wave is 30 W/m². What is the peak value of the Electric field in the wave? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C²/Nm²)

$I = c\,\varepsilon_o E_{rms}^2 \quad | \quad E_{rms} = \sqrt{\dfrac{I}{c\varepsilon_o}} = 106.3$ N/C $\quad | \quad E_p = E_{rms}(\sqrt{2})$

$E_p = 150.3$ N/C

13) What area of solar cells would be needed to power a house that uses 5,000 W of power and the sun's intensity is 1,000 W/m² if the solar cells are 15 % efficient at converting the suns energy to electrical energy?

$I = \dfrac{P}{A} = 0.15(I) \quad | \quad A = \dfrac{P}{0.15(I)} = 33.3$ m²

14) The intensity of the electromagnetic wave has an average value of 60 W/m^2. The wave is incident at 0° to the normal on a rectangular solar cell with an area measuring 1.5 m by 2.0 m. How much total energy falls upon the solar cell each minute?

$I = \frac{P}{A} = \frac{(E/\Delta t)}{A}$ | $\Delta t = 60s$ | $E = IA\Delta t =$ 10,800 J

15) If a solar cell with an area of 2 m^2 receives 800 W/m^2 of light Intensity when the normal to the surface is at 0° to the sun, how much power does it receive when the sun is at an angle of 50° measured normal to the surface of the cell?

$P = IA\cos\phi$ | $\phi = 50°$ | $P = (800 W/m^2)(2 m^2)\cos(50°)$

$P =$ 1,028 W

16) A 60.0 W light bulb radiates light uniformly in all different directions. What is the average intensity of the light bulb at a distance of 0.400 m from the bulb? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2)

$I = \frac{P}{4\pi r^2} =$ 29.8 W/m^2

17) A Tree's canopy has a cross sectional area of 50 m^2, how much energy does the tree receive from the sun in 1 hour if the sun's intensity is 800 W/m^2?

$I = \frac{P}{A} = \frac{(E/\Delta t)}{A}$ | 1 hr = 3600 s | $E = IA\Delta t =$ 1.44×10^8 J

18) A 60.0 W light bulb radiates light uniformly in all different directions. What is the rms value of E at a distance of 0.400 m from the bulb? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2)

$I = \frac{P}{4\pi r^2} = \varepsilon_o c E^2$ | Solve For E | $E = \sqrt{\frac{P}{4\pi r^2 \varepsilon_o c}} =$ 106 N/C

19) An 8.000-mW laser beam emits a cylindrical beam of light 0.9000 mm in diameter. What is the rms value of the electric field in this beam? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2)

$A = \pi r^2 = 6.4 \times 10^{-7}$ m^2

$I = \varepsilon_o c E^2 = \frac{P}{A}$ | Solve For E | $E = \sqrt{\frac{P}{\varepsilon_o c A}} =$ 2,170 N/C

20) For a 2,000 W radio transmission tower, what is the intensity of the radio waves 2 km from the tower? ($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2)

$I = \frac{P}{4\pi r^2} = \frac{2{,}000 W}{4\pi(2{,}000 m)^2} =$ 3.98×10^{-5} W/m^2

r = 2,000 m

P = 2,000 W

21) Cathy measures the electric field strength (rms) of a radio transmission tower at her house to be 0.12 N/C, if the tower produces 10,000 W EM radio wave power, how far is Cathy from the tower?
($c = 3 \times 10^8$ m/s $\mu_o = 1.257 \times 10^{-6}$ Tm/A $\varepsilon_o = 8.85 \times 10^{-12}$ C²/Nm²)

$I = \frac{P}{4\pi r^2} = \varepsilon_o c E^2$ Solve for r | $r = \sqrt{\frac{P}{4\pi \varepsilon_o c E^2}} = 4,562\text{ m}$

22) The temperature of lava flow is 1,200 °C, if the area of the lava is 2 m², the emissivity is e = 0.95 and the background temperature is 30 °C, what is the net power of the EM waves being emitted by the lava.

$P_{net} = e\sigma A(T^4 - T_o^4) = 5.1 \times 10^5\text{ W}$

$e = 0.95$ | $\sigma = 5.67 \times 10^{-8}\text{ J/m}^2\text{K}^4$ | $T = 1,200°C + 273° = 1473\text{K}$ | $T_o = 30°C + 273° = 303\text{K}$ | $A = 2\text{m}^2$

23) If a glass window with an area of 1 m2 and an emissivity of 0.9 is at a temperature of 30 °C, the outside temperature is 0°C, how much power is being lost through the window?

$A = 1\text{m}^2$ | $\sigma = 5.67 \times 10^{-8}\text{ J/m}^2\text{K}^4$ | $e = 0.9$ | $T = 30°C + 273° = 303\text{K}$ | $T_o = 0°C + 273° = 273°$

$P_{net} = e\sigma A(T^4 - T_o^4)$

$P_{net} = 147\text{ W}$

24) The heating element in Bob's toaster oven has an area of 0.01 m², an emissivity of 0.9 and is in air with a background temperature of 25 °C. If the EM waves power being emitted by the heating element is 3,000 W, what is the temperature of the element? (1 nm = 10^{-9}m) $T_o = 25°C + 273 = 298\text{K}$

$P = e\sigma A(T^4 - T_o^4)$

$T^4 - T_o^4 = \frac{P}{e\sigma A}$ → $T^4 = \frac{P}{e\sigma A} + T_o^4$

$T = \sqrt[4]{\frac{P}{e\sigma A} + T_o^4} = 1,518\text{K}$ $1,245°C$

25) The temperature of lava flow is 1,200 °C, what is the peak wavelength emitted by the lava?

$\lambda_p = \frac{2.898 \times 10^{-3}\text{ mK}}{T} = \frac{2.898 \times 10^{-3}\text{ mK}}{1473\text{K}} = 1.97 \times 10^{-6}\text{ m}$

$1.97\,\mu\text{m}$

26) The heating element in Bob's toaster oven emits electromagnetic waves with a peak wavelength of 650 nm, what is the temperature of the heating element? (1 nm = 10^{-9}m)

$\lambda_p = \frac{(2.898 \times 10^{-3}\text{ mK})}{T}$ | $T = \frac{2.898 \times 10^{-3}\text{ mK}}{\lambda_p} = 4,458\text{K}$

$4,185°C$

27) If the color temperature of a light bulb is 2,500 K, what type of light do you see?

A) Bright white light.
B) Whiter light.
C) Yellowed light.

28) What is a good estimate of the Intensity of EM waves at the surface of the earth from the sun on a clear day?

A) 2,000 W/m^2
B) 100 W/m^2
C) 1,370 W/m^2
D) 1,000 W/m^2

29) The peak wavelength emitted by the sun is closest to

A) 450 nm
B) 600 nm
C) 650 nm
D) 550 nm
E) 340 nm

30) The majority of the sun emits in what area of the EM spectrum?

A) Infrared radiation
B) Visible radiation
C) Ultraviolet radiation
D) Microwave radiation
E) x-ray and gamma ray radiation

KEY

1) B
2) A
3) A
4) C
5) B
6) 1.9×10^{-6} T
7) 1,500 N/C
8) 863 W/m^2
9) 5,970 W/m^2
10) 3.54×10^{-7} T
11) 5×10^{-7} T
12) 150.3 N/C
13) 33.3 m^2
14) 10,800 J
15) 1,028 W
16) 29.8 W/m^2
17) 1.4×10^{8} J
18) 106 N/C
19) 2,177 N/C
20) 3.98×10^{-5} W/m^2
21) 4,562 m
22) 5.1×10^{5} W
23) 147 W
24) 1,558 K
25) 1.97×10^{-6} m
26) 4,458 K
27) C
28) D
29) D
30) A